

THE BEST THING A HUMAN BEING
CAN DO IS TO HELP ANOTHER
HUMAN BEING KNOW MORE.

- CHARLIE MUNGER



Agenda

- ① Map vs FlatMap in RDD
- ② Jobs vs Stages vs Tasks
- ③ Coding with RDD in movies Dataset

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- ① Introduction to spark SQL module
 - ② components in spark SQL module
 - ③ using SQL in spark session
 - ④ DSL (DataFrames) Demo

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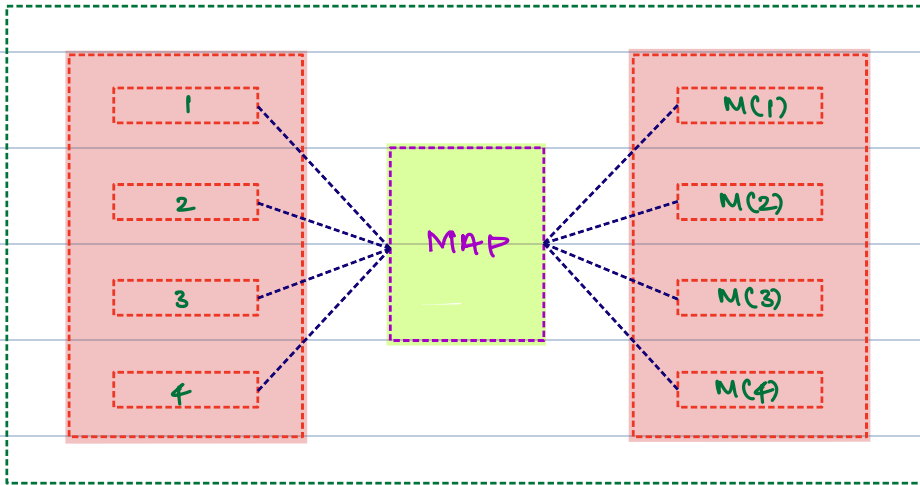
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Quick update : Pager Duty

RDD operations

Flatmap vs map in spark RDD

Map: Transforms one input element to one output element

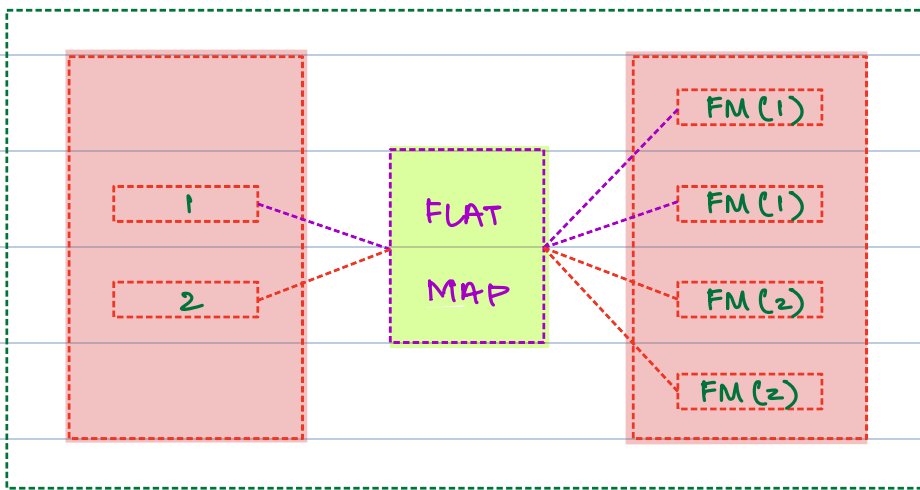


$a = [1, 2, 3, 4, 5]$

square()

output: $[1, 4, 9, 16, 25]$

Flatmap: Transform one input element into zero or more output element.



$a = ["Suraj Adhule", "Nandeep Raj"]$

flat map $\rightarrow$ split()

Suraj Adhule $\rightarrow$ $["Suraj", "Adhule"]$

Nandeep Raj $\rightarrow$ $["Nandeep", "Raj"]$

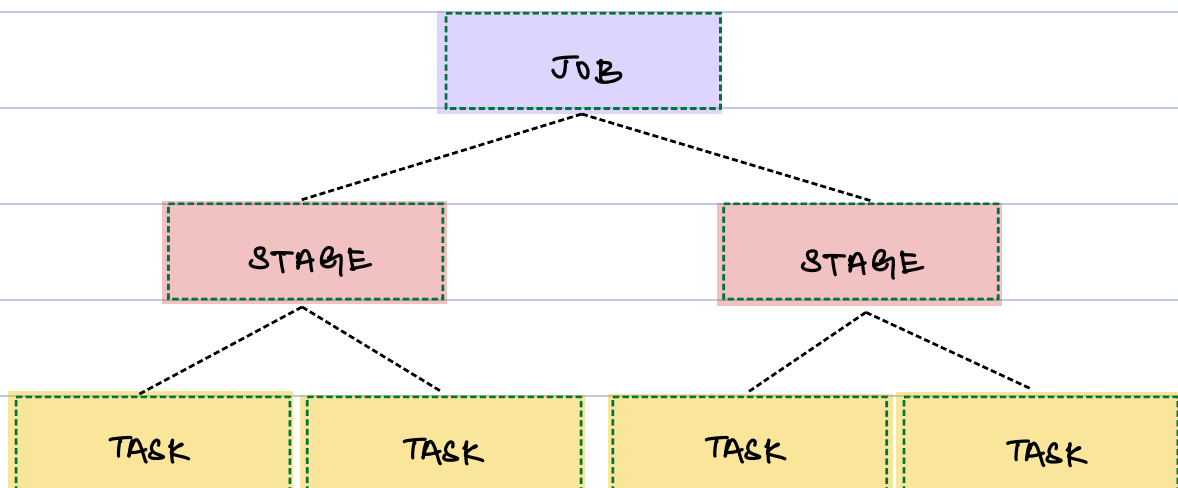
Jobs vs Stages vs Tasks

Job: created whenever an action is performed to RDD or other spark object.

Stage: Created when a wide transformation needs to be applied in a Job

Task: The smallest unit of work that needs to be performed in a single partition in a stage.
The number of tasks in the 1st stage will be equal to number of partition in source data

Hierarchy



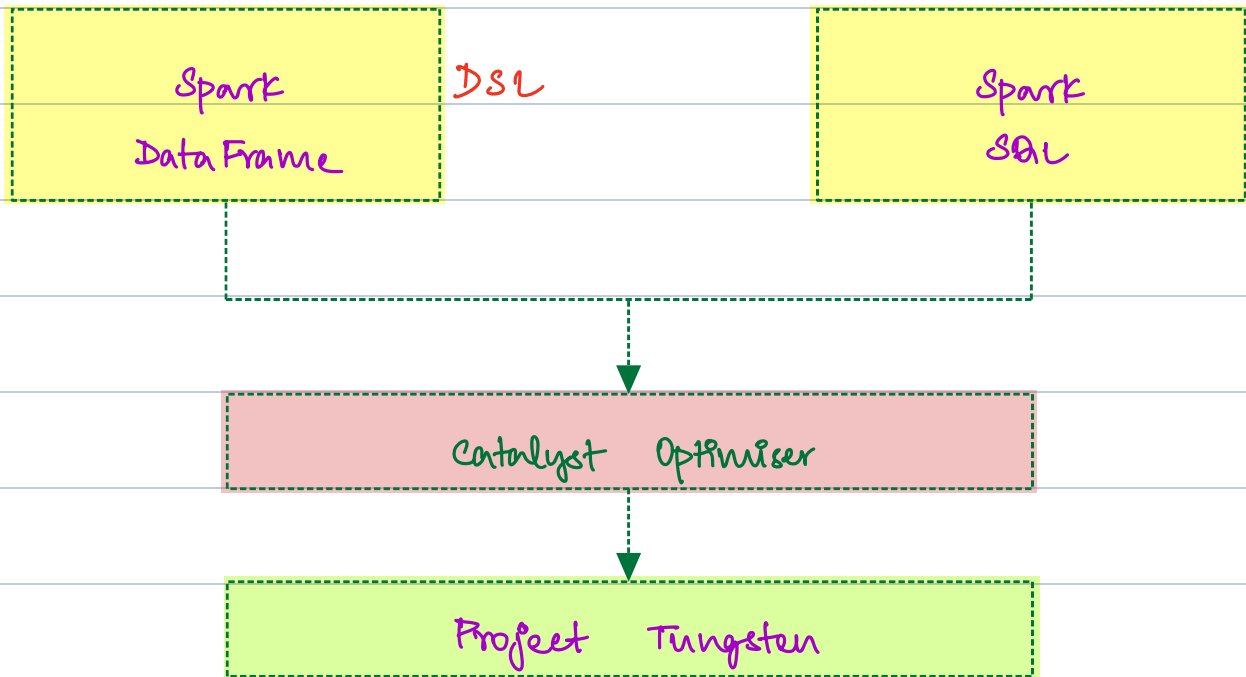
RDD In Demo

- ✓ ① Create Spark Context
- ✓ ② Read data from movies and ratings
- ✓ ③ Filter for only comedy movies
- ✓ ④ Join movies and ratings based on movie ID
- ⑤ Group by movie ID or movie name and count the number of ratings per movie
- ⑥ Get movie name with max count

10:13 pm → 10:20 pm

mId	title	rating id	movie id	fs	
1	abc	5	1	2	→ row 1
1	abc	6	1	3	→ row 2
2	def	7	2	4	→ row 3
2	def	8	2	5	→ row 4
2	def	9	2	6	→ row 5

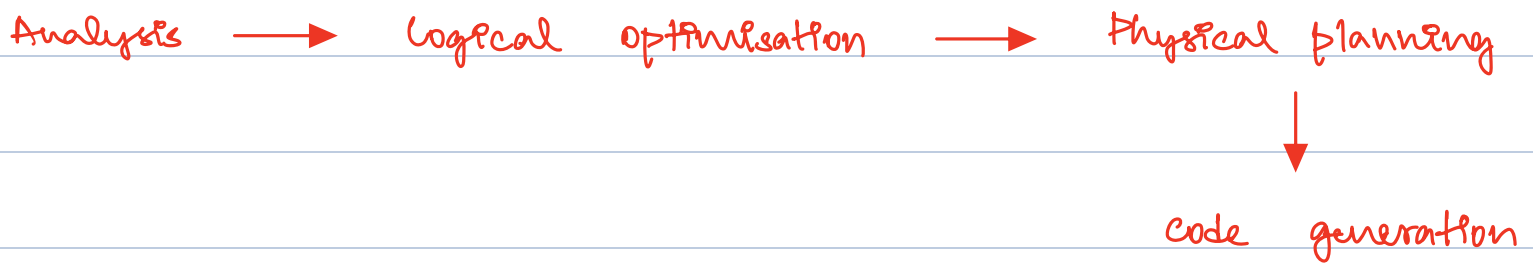
Spark SQL



Spark SQL is a highlevel API that provides abstraction on top of your RDB. You can write your business logic using Domain specific language (DSL) or SQL which provides all essential function.

Catalyst Optimiser

Heart of spark SQL that provides performance enhancement. The catalyst optimiser processes query through sequence of stages to produce highly efficient physical plan.



Predicate Pushdown : where

Projection Pruning : SELECT columns

Logical

optimisation

Cost Based Optimisation : cost model

Physical Planning

Project Tungsten Execution Engine

Tungsten is an Intelligent Execution Engine. It has three pillars that optimises the Execution.

Off-Heap memory utilization. (avoids overload in JVM)

Better Caching Plan : Uses different Datatype Closes to CPU caching

Better serialization of data : Introduced better serialization & deserialization designs (Unsafe Row)